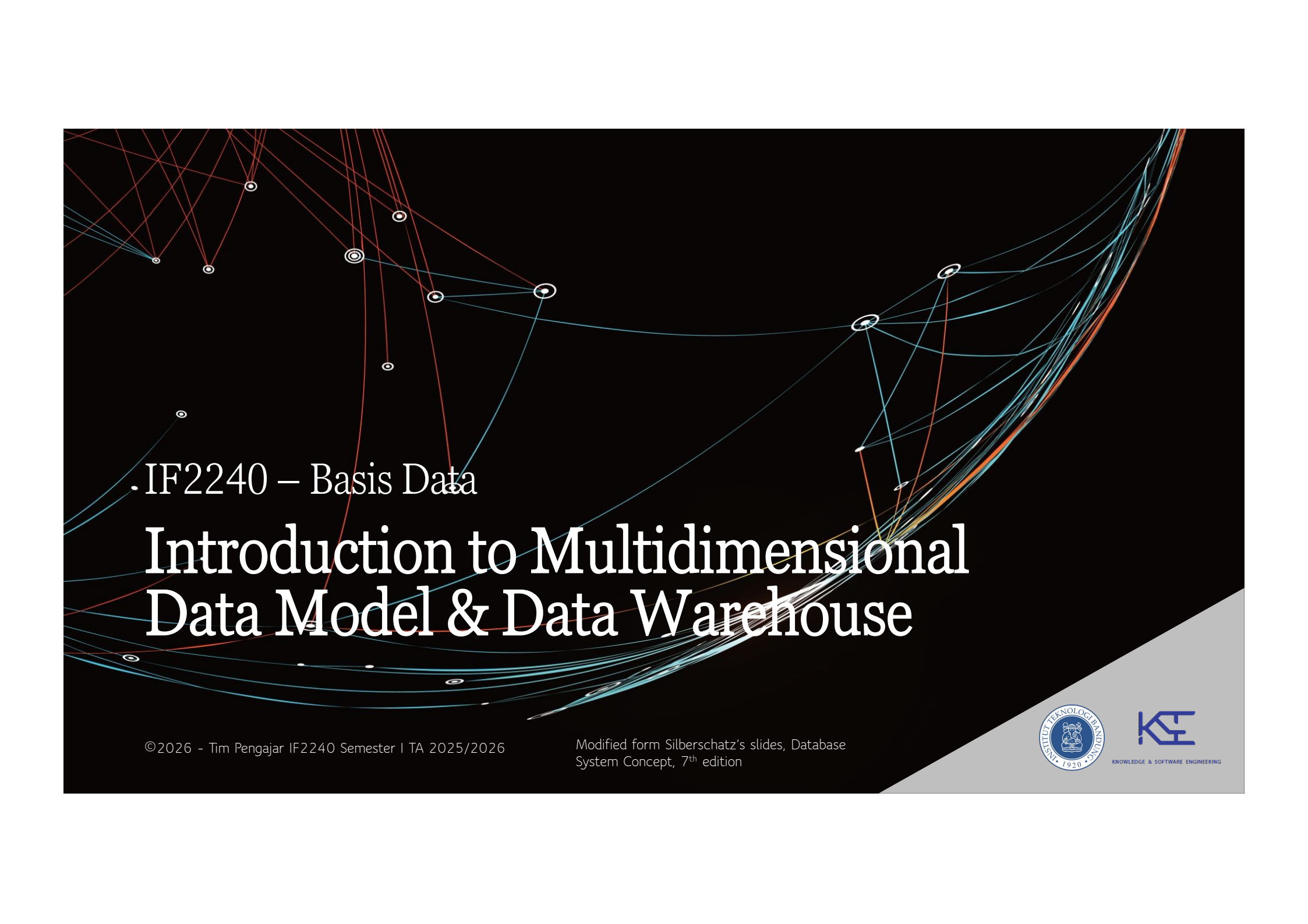




. IF2240 – Basis Data

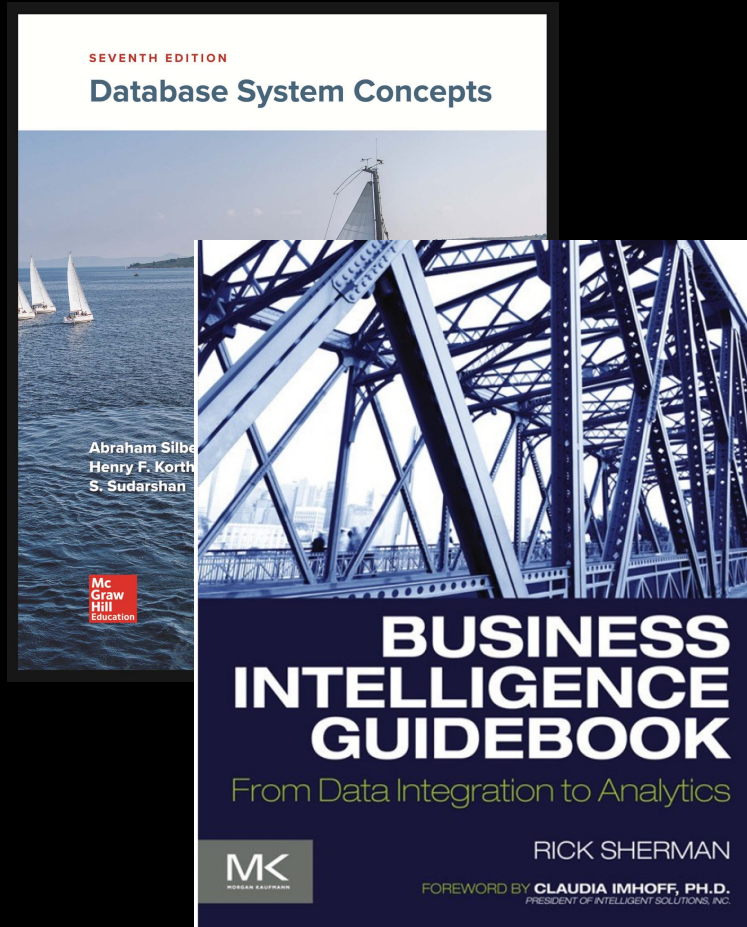
Introduction to Multidimensional Data Model & Data Warehouse

©2026 - Tim Pengajar IF2240 Semester I TA 2025/2026

Modified from Silberschatz's slides, Database
System Concept, 7th edition



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References

Abraham Silberschatz, Henry F. Korth, S. Sudarshan :
“Database System Concepts”, 7th Edition

- Chapter 11: Data Analytics

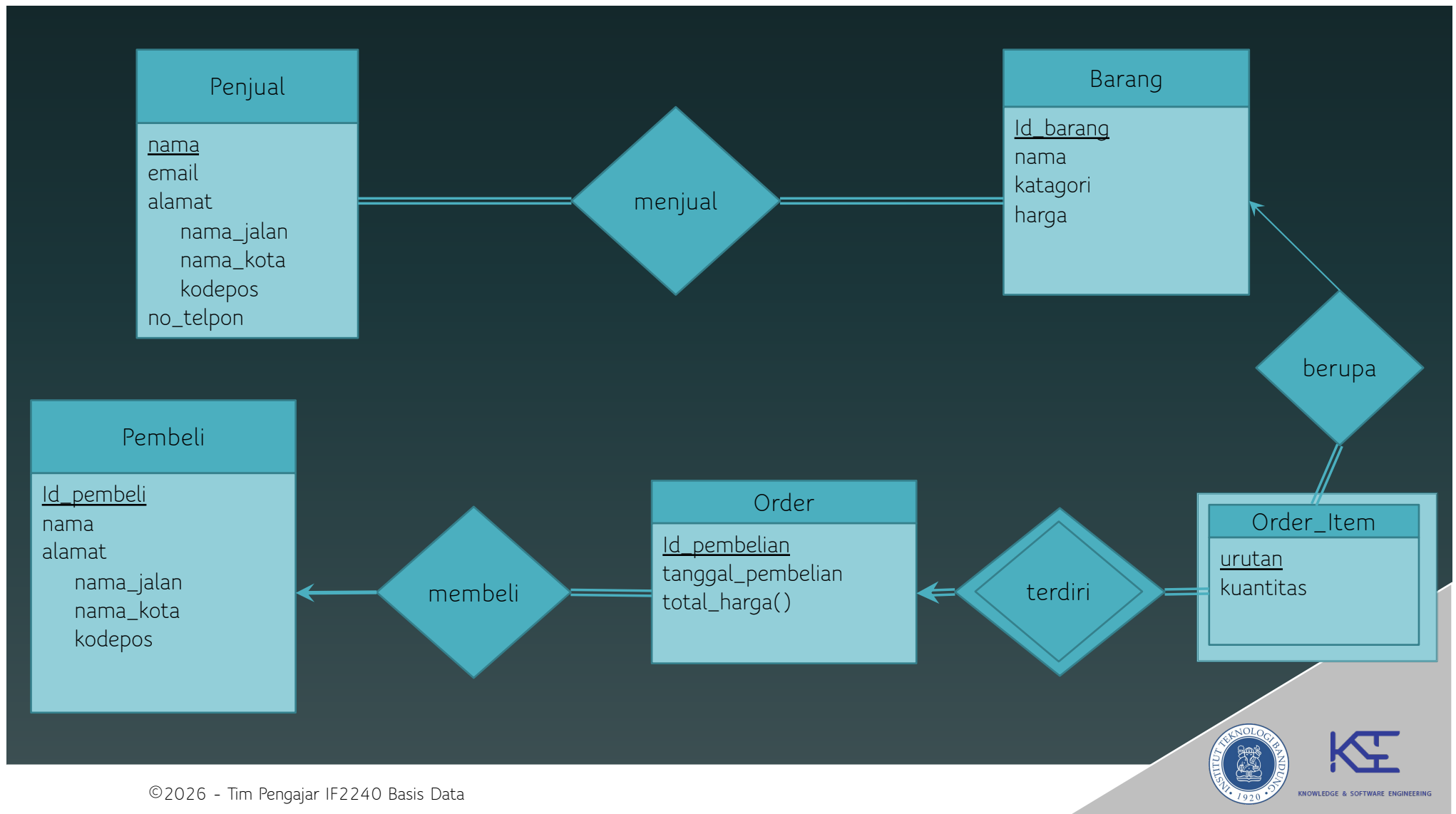
Rich Sherman: “Business Intelligence Guidebook : From Data Integration to Analytics”, 1st Edition

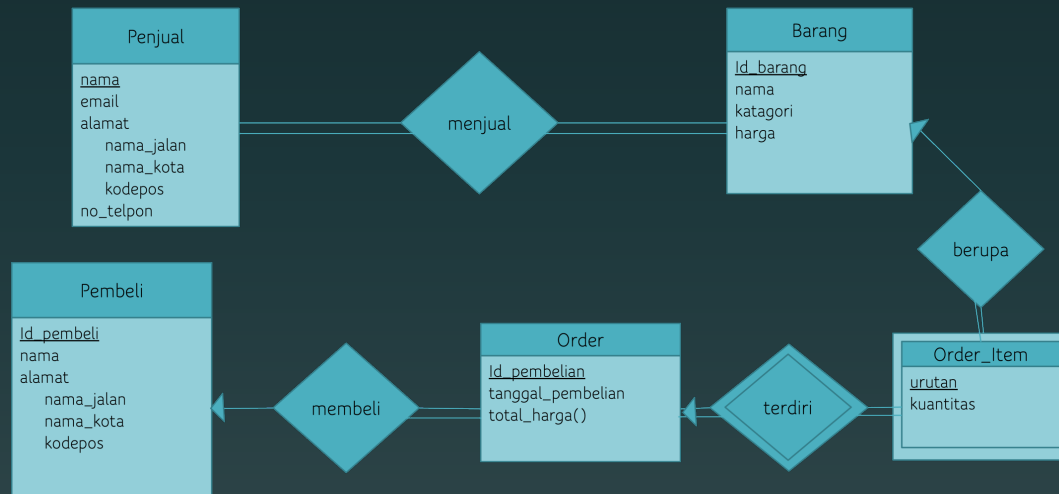
- Chapter 1: The Business Demand for Data, Information, and Analytics

Pemodelan untuk Toko Online

Itsy Bitsy adalah perusahaan e-commerce yang menjual berbagai barang dari berbagai penjual. Saat ini mereka tengah membangun *database* yang akan membantu mereka dalam mencatat **penjual** dan masing-masing **barang** yang mereka jual. Masing-masing penjual memiliki nama yang unik, kontak email, alamat, dan nomor telepon. Penjual dapat **menjual satu atau lebih barang**, masing-masing jenis barang memiliki id yang unik, nama, katagori dan harga.

Pengembangan lebih lanjut, dilakukan penyimpanan informasi **pembeli**, berupa id, nama dan alamat (terdiri atas nama jalan, nama kota, dan kodepos). Setiap pembeli dapat melakukan **pembelian**. Satu kali pembelian memiliki id, tanggal pembelian, dan total harga yang dijumlahkan dari harga masing-masing barang yang dibeli. Setiap jenis barang yang dibeli perlu dicatat kuantitasnya.





Penjual : Nama, email, alamat, alamat_nama_jalan, alamat_nama_kota, alamat_kodepos, no_telpon

Barang : id_barang, nama, katagori, harga

Menjual : nama_penjual, id_barang

Order : id_pembelian, tanggal_pembelian, id_pembeli

Order_item : id_pembelian, urutan, kuantitas, id_barang

Pembeli : id_pembeli, nama, alamat_nama_jalan, alamat_nama_kota, alamat_kodepos

FK

Menjual(nama_penjual) → Penjual (nama)

Menjual(id_barang) → Barang(id_barang)

Order(id_pembeli) → Pembeli(id_pembeli)

Order_item(id_pembelian) → Order(id_pembelian)

Order_item(id_barang) → Barang (id_barang)

Kebutuhan Informasi untuk Tim Eksekutif (Analysis)

1. Berapa total penjualan barang setiap bulannya?
2. Berapa rata-rata jumlah rupiah penjualan setiap bulannya?
3. Penjual mana yang menjual barang terbanyak bulan ini?
4. Di kota mana market terbesar dari perusahaan?
5. Barang apa yang paling laku terjual?

Contoh masalah yang akan ditimbulkan jika menggunakan model relasional:

1. Peningkatan kompleksitas query
2. Mungkin memerlukan waktu eksekusi yang lama dan mengganggu proses operasional

Sebuah ilustrasi... dari skema BD lain..

Siapa Penjual Top Skin Care di 2023?

```
1  SELECT
2      s.email
3      SUM(o.total_price) AS revenue,
4      SUM(oi.quantity) AS product_sold
5  FROM order AS o
6      LEFT JOIN order_item AS oi ON o.id = oi.order_id
7      LEFT JOIN product AS p ON oi.product_id = p.id
8      LEFT JOIN product_category AS pc ON p.category_id = pc.id
9      LEFT JOIN seller AS s ON o.seller_id = s.id
10 WHERE o.order_date BETWEEN '2023-01-01' AND '2023-12-31'
11      AND pc.name = "Skin Care"
12 GROUP BY s.email
13 ORDER BY revenue DESC
14 LIMIT 5
```

Contoh masalah yang akan ditimbulkan jika menggunakan model relasional:
Peningkatan kompleksitas query

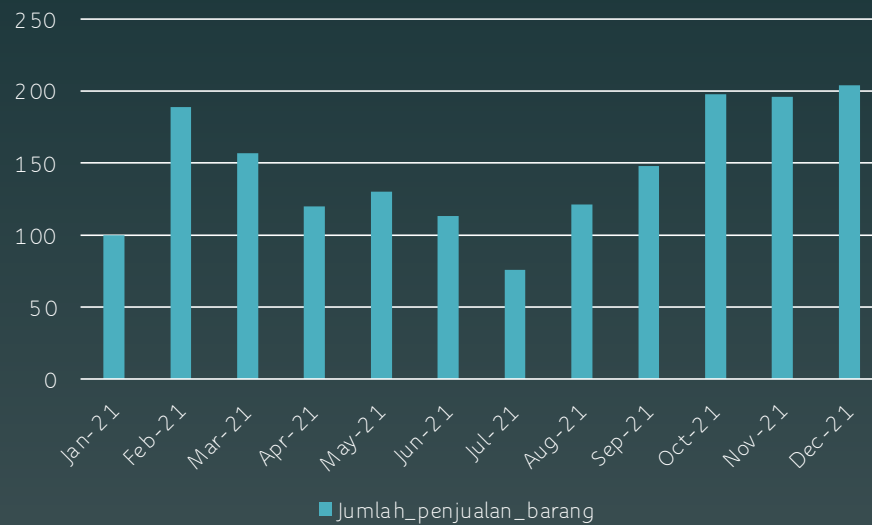
Kebutuhan Informasi Pihak Eksekutif digunakan untuk keperluan analisis sehingga melibatkan data dalam jumlah yang besar...

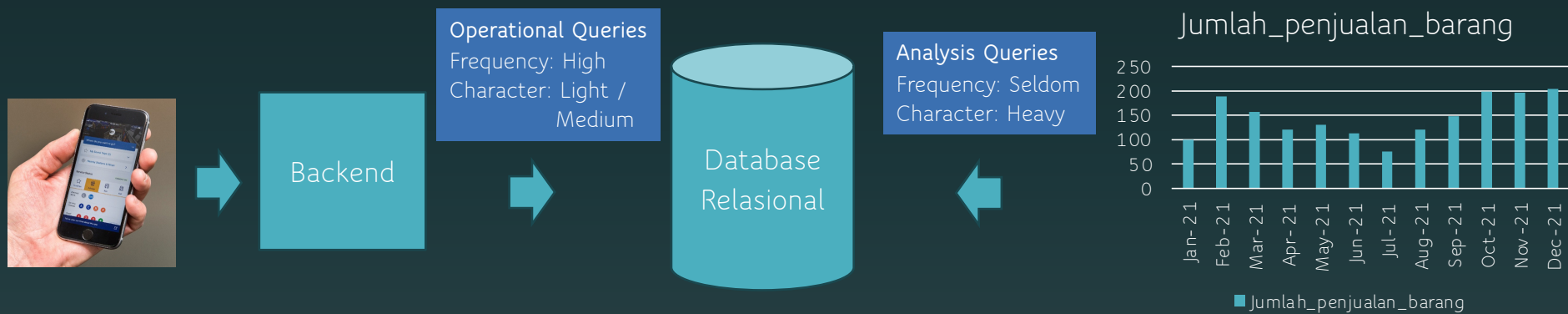
Filter jenis barang : semua ▼

Filter jenis cuaca : cerah ▼

Filter jenis penjual : semua ▼

Jumlah_penjualan_barang





Contoh masalah yang akan ditimbulkan jika menggunakan model relasional:
Mungkin memerlukan waktu eksekusi yang lama dan **mengganggu proses operasional**

Multidimensional (1/2)

The multidimensional model begins with the observation that the factors affecting decision-making processes are **enterprise-specific facts**, such as sales, shipments, hospital admissions, surgeries, and so on. Instances of a fact **correspond to events that occurred**. For example, every single sale or shipment carried out is an event. Each fact is described by the values of a set of **relevant measures that provide a quantitative description of events**. For example, sales receipts, amounts shipped, hospital admission costs, and surgery time are measures.

Terminology

- **Dimension**: subject label for a row or column
- **Member**: value of dimension
- **Measure**: quantitative variables stored in cells

Multidimensional (2/2)

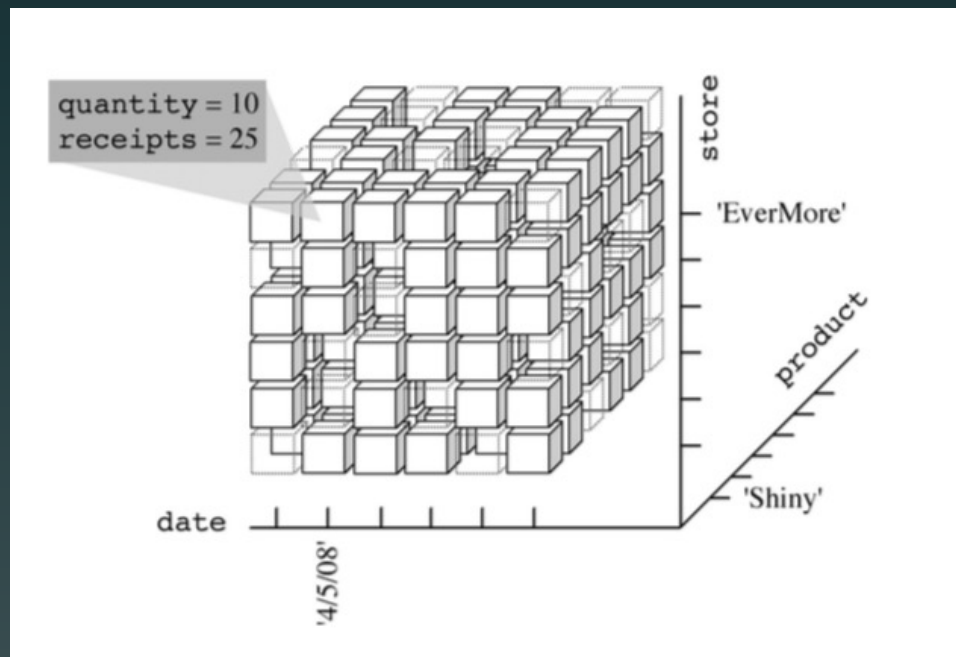
Used metaphor of *cubes* to represent multidimensional data.

Events are associated with **cube cells**. Each cube cell is given a value for each measure

Cube edges stand for analysis dimensions.

If more than three dimensions exist, the cube is called a *hypercube*.

Sales Data Cube Example



The three-dimensional cube modeling sales in a store chain:
10 packs of Shiny were sold on 4/5/2008 in the EverMore store, totaling \$25

Relational schema :

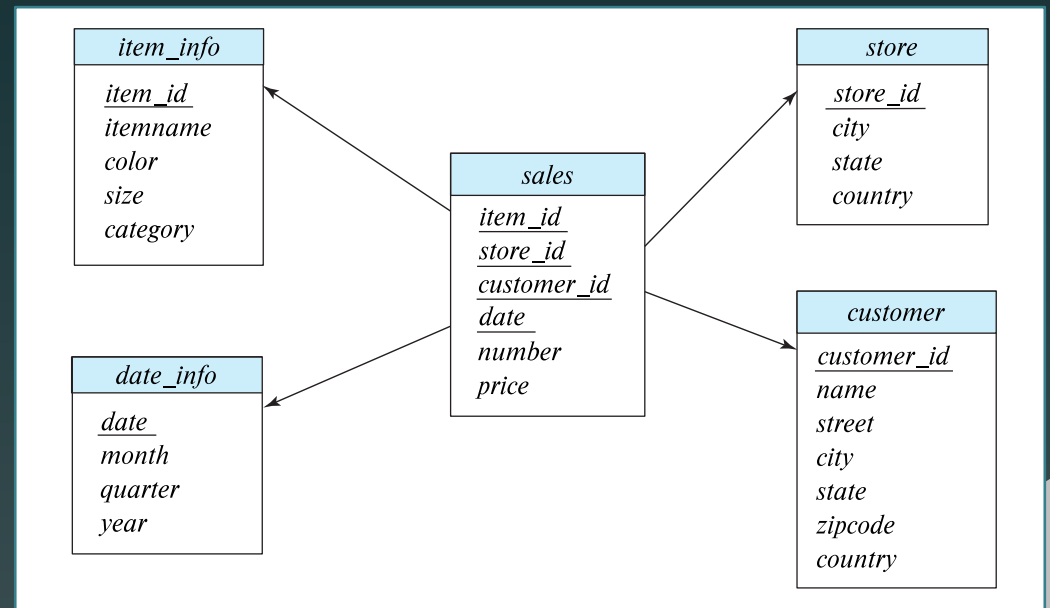
SALES(store, product, date,
quantity, receipts)

<'EverMore', 'Shiny', '04/05/08', 10, 25>

Multidimensional Data and Warehouse Schemas (1/2)

Data in warehouse can usually be divided into

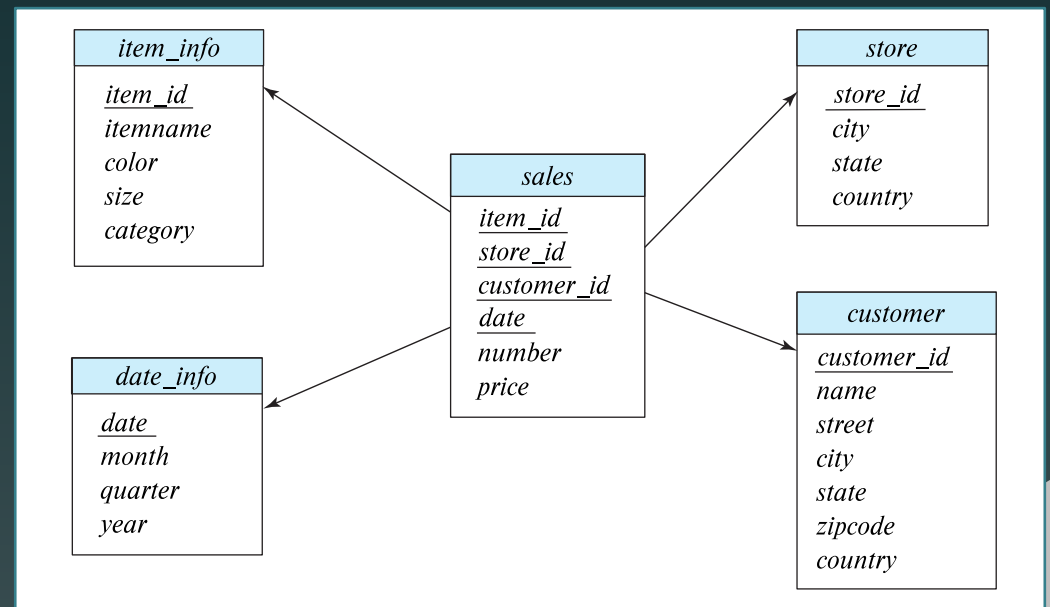
- **Fact tables**, which are large
 - E.g, *sales(item_id, store_id, customer_id, date, number, price)*
- **Dimension tables**, which are relatively small
 - Store extra information about stores, items, etc.



Multidimensional Data and Warehouse Schemas (2/2)

Attributes of fact tables can be usually viewed as

- **Measure attributes**
 - measure some value, and can be aggregated, e.g., the attributes *number* or *price* of the *sales* relation
- **Dimension attributes**
 - dimensions on which measure attributes are viewed, e.g., attributes *item_id*, *color*, and *size* of the *sales* relation
 - Usually small ids that are foreign keys to dimension tables



Kebutuhan Informasi untuk Tim Eksekutif

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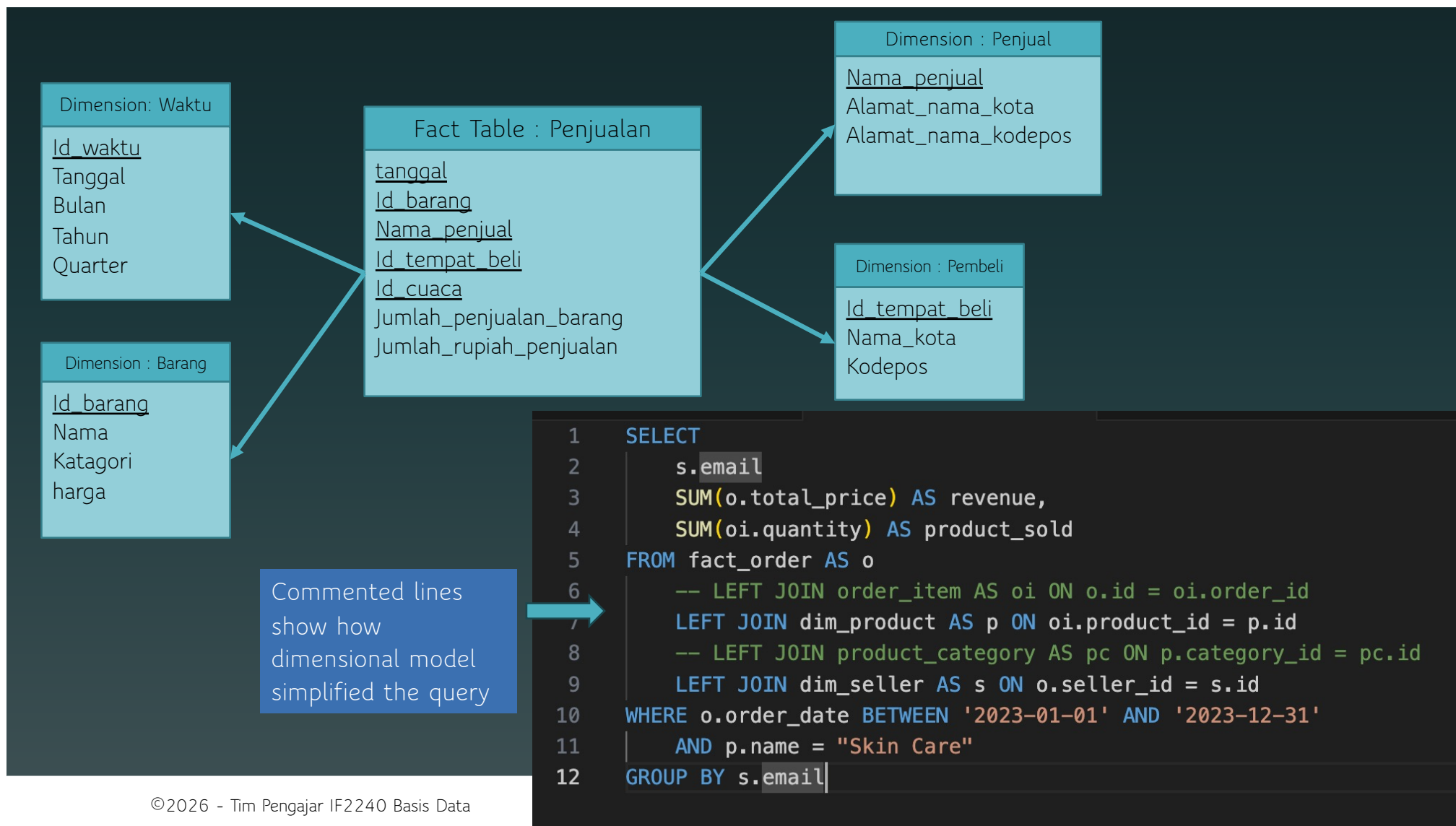
Dapat diselesaikan dengan pendekatan model multidimensional



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Adanya kebutuhan dari pihak manajerial untuk mengetahui trend penjualan dan faktor yang mempengaruhi trend tersebut.





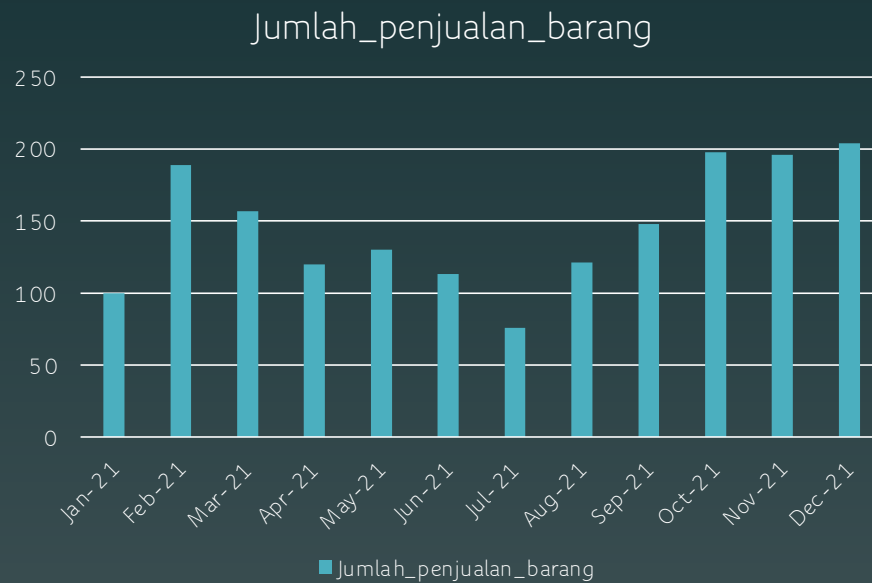
Multidimensional model tersebut dapat membantu meng-generate report untuk analisis trend yang dibutuhkan:

Filter jenis barang : semua ▼

Filter jenis cuaca : cerah ▼

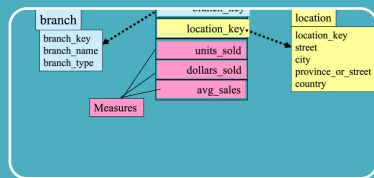
Filter jenis penjual : semua ▼

Bisa menambahkan filter sesuai dengan dimensi yang tersedia



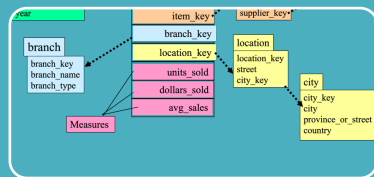
Conceptual Modeling of Data Warehouses

Modeling data warehouses: dimensions & measures



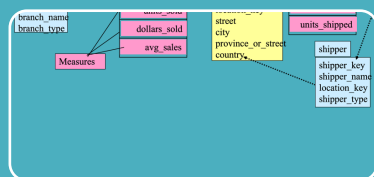
Star schema:

- A fact table in the middle connected to a set of dimension tables



Snowflake schema:

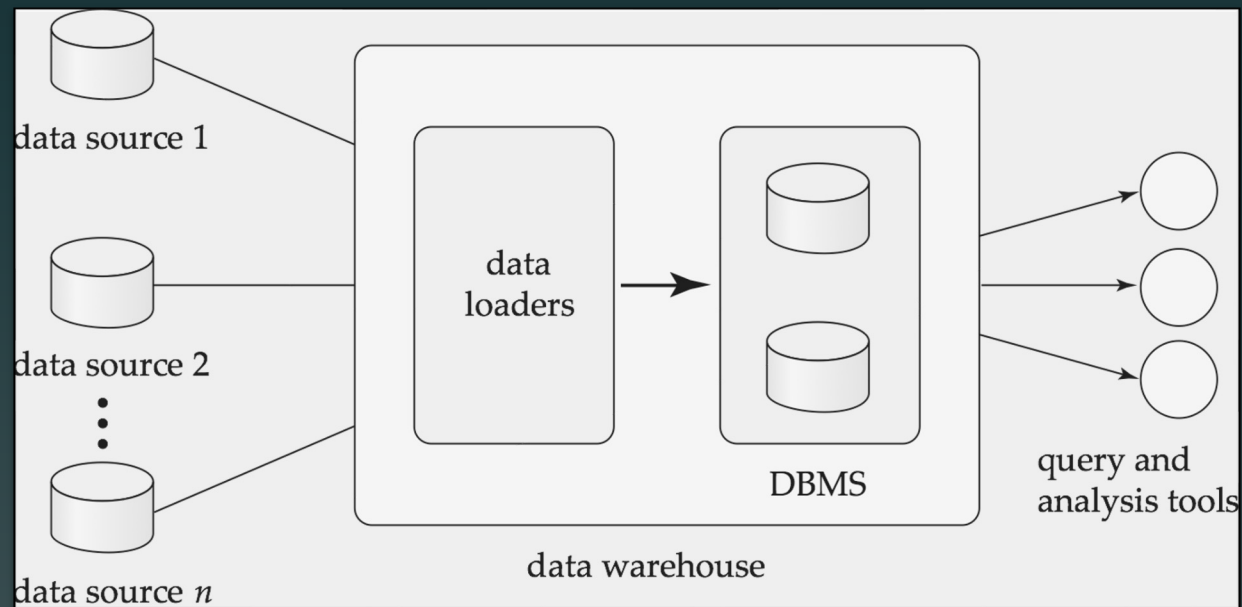
- A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake



Fact constellations:

- Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

Data Warehousing



Data Warehousing

Data sources often store only current data, not historical data

Corporate decision making requires a unified view of all organizational data, including historical data

A **data warehouse** is a repository (archive) of information gathered from multiple sources, stored under a unified schema, at a single site

- Greatly simplifies querying, permits study of historical trends
- Shifts decision support query load away from transaction processing system



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Data Warehouse vs. Operational DBMS #01

OLTP (on-line transaction processing)

Major task of traditional relational DBMS

Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.

OLAP (on-line analytical processing)

Major task of data warehouse system

Data analysis and decision making



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Data Warehouse vs. Operational DBMS #02

Distinct features (OLTP vs. OLAP):



User and system orientation: customer vs. market



Data contents: current, detailed vs. historical, consolidated



Database design: ER + application vs. star + subject



View: current, local vs. evolutionary, integrated



Access patterns: update vs. read-only but complex queries



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OLTP vs. OLAP

	OLTP	OLAP
Users	Clerk, IT Professional	Knowledge worker
Function	Day to day operations	Decision support
DB Design	Application-oriented	Subject-oriented
Data	Current, up-to-date Detailed, flat relational Isolated	Historical Summarized, multi-dimensional Integrated, consolidated
Usage	Repetitive	Ad-hoc
Access	Read/Write Index/hash on primary key	Lots of scans
Unit of Work	Short, simple transaction	Complex query
# Records Accessed	Tens	Millions
# Users	Thousands	Hundreds
DB Size	100MB-GB	100GB-TB
Metric	Transaction throughput	Query throughput, response

